

Dr Joshua Mark Pollin

Thermonuclear Supernovae: From 3D Explosion Models to Nebular Observables

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Education & Affiliations

Oregon State University , USA – Postdoctoral Researcher in Astrophysics. Department of Physics; group of Dr Sanjana Curtiss.	Jan 2026 – Present
Queen's University Belfast , UK – PhD Astrophysics. Thesis: <i>Exploring the Origins of Type Ia Supernovae with Monte Carlo Radiative Transfer</i> . Advisors: Prof. Stuart Sim and Dr Matt Nicholl. Thesis submitted: 19 Sep 2025 (2 years 11 months since PhD Start). PhD awarded: 10 Dec 2025.	Oct 2022 – Dec 2025
Queen's University Belfast , UK – Integrated MSci Physics (First Class Honours). Advisor: Prof. Stuart Sim.	Sep 2018 – Jul 2022

Skills & Expertise

Simulation & Modelling: 3D time-dependent NLTE Monte Carlo radiative transfer (ARTIS); core-collapse supernova modelling (PUSH); nucleosynthesis (CFNET).

Software Development: ARTIS development enabling fully 3D nebular-phase NLTE calculations; ARTISTOOLS development for multidimensional analysis and visualisation.

HPC & Computing: C/C++, Python, Fortran, Bash, MPI, OpenMP; Git and CI/CD; management of simulation workflows using ~7M CPU hours per year.

Grants & HPC Proposals

ESO VLT FORS2 – 4 h service-mode time (P117) Co-Investigator: <i>Nebular Spectroscopy of Long-Duration Stripped-Envelope Supernovae</i> (PI: Thomas Moore). <i>Five successful HPC proposals totalling ~95M CPU hours; one as sole science-case lead.</i>	May 2026 – Apr 2027
SuperMUC-NG (LRZ), Germany – 35M CPU hours Co-authored proposal; led ~10% of the science case. Outputs: 1 first-author paper in review and 1 co-authored publication.	Mar 2025 – Jan 2027
DiRAC HPC Facility, UK – ~36M CPU hours Co-authored proposal; led ~50% of the science case. Outputs: 2 first-author and 2 co-authored publications.	Apr 2024 – Mar 2027
JUWELS (JSC/HITS), Germany – 21M CPU hours Co-authored two successive proposals; led ~30% of the science case. Outputs: 2 co-authored publications.	May 2023 – Apr 2025
DiRAC HPC Facility, UK – 3M CPU hours Sole science-case lead (100%). Output: 1 first-author publication.	Apr 2023 – Mar 2024

Scholarships & Awards

O'Sullivan Masterclass , Royal Irish Academy – selected as a participant for a Nobel laureate masterclass.	Feb 2025
Student Research Grant , QUB – led a successful competitive application for a summer student research project; ~£2,000 awarded.	Jun – Aug 2024
Emily S. Montgomery Travel Scholarship – £400 competitive travel award.	Apr 2024

Invited Talks, Conferences & Organisation

[INVITED]	Progenitors of Thermonuclear Supernovae , Leiden; upcoming	Jan 2027
[INVITED]	CIERA, Northwestern University , Evanston, USA; research visit and talk	Aug 2026
[SEMINAR]	Trinity College Dublin , Ireland	Mar 2025
[ORGANISED]	National Astronomy Meeting , Durham; co-proposed, co-organised and co-chaired <i>Explosive Transients in the Present and Future Sky</i> (~50–60 participants)	Jul 2025
[Contributed]	Rise_Time 2026, Purdue University, USA	Aug 2026
[Contributed]	16th–19th Würzburg Winter Workshops, Germany	Dec 2022–2025
[Contributed]	RAS Specialist Discussion Meeting, London	Nov 2024
[Contributed]	Oxford Transients Meeting	Jul 2024
[Contributed]	European Workshop on White Dwarfs, Barcelona	Jul 2024
[Contributed]	National Astronomy Meeting, Cardiff	Jul 2023

Student Mentoring

Lindsey Wingate , OSU – assist in supervising a senior thesis on CCSN nucleosynthesis; research meetings, code support, and review of analysis and writing.	Aug 2026 – Present
Adam McGarrity , QUB – PhD mentoring, HPC training and ARTIS support; regular research meetings, continuing virtually since leaving QUB.	Oct 2025 – Present
Fiona McNeil , QUB – PhD mentoring and training in ARTIS workflows and analysis.	Oct 2023 – Jan 2026
Anna Hayes , QUB – co-mentored a summer project on microphysics in ARTIS; subsequently progressed to a PhD programme.	Summer 2024
Ronan Agnew , QUB – MSc project mentoring on kilonovae.	Oct – Dec 2023

Teaching, Outreach & Public Engagement

Graduate Teaching Assistant , QUB – mathematics tutorials; Teaching Fellowship Scheme.	Sep 2023 – Dec 2025
DiRAC Science Highlights – co-authored five supernova features, including two on my own research.	May 2023–2026
Work Experience Mentoring , QUB – school-student mentoring, research seminars and computational modelling activities.	Jul 2025
Girls in Maths & Physics , QUB – annual astrophysics talks and interactive exoplanet workshops.	Jun 2023–2025
Aurora Outreach Video – co-produced a public-engagement video (~150,000 views).	May 2024
NI Science Festival – public lecture on using light to probe the Universe.	Feb 2024
Irish Astronomical Association – invited public talk on Type Ia supernovae.	Jan 2024
Pathway Opportunity Programme , QUB – developed and delivered computational workshops on cosmic expansion.	2023–2024
Undergraduate Peer Mentor , QUB – supported incoming physics students.	Sep 2019 – May 2020

References

Prof. Stuart Sim
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Dr Sanjana Curtis
Oregon State University
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Prof. Friedrich Röpke
HITS Heidelberg
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Publications

Since beginning full-time research in October 2022, I have published 9 journal papers (3 as first author). A further 2 papers are under review (1 as first author), and 1 first-author manuscript has completed collaboration review. Google Scholar: 113 citations including software; h-index 6 (October 2026).

First author	Citations	Year
1. J. M. Pollin et al. <i>The Impact of Binary Stripping of Massive Stars on Core-Collapse Supernovae – I. Explodability and Nucleosynthesis</i> . Collaboration review complete; to be submitted to MNRAS.	[-]	2026
2. J. M. Pollin et al. <i>Nebular Fingerprints of a Violent White Dwarf Merger: 3D NLTE Modelling of Type Ia Supernovae</i> . MNRAS, in review.	[-]	2026
3. J. M. Pollin et al. <i>Updated 3D NLTE Nebular Diagnostics Favor D6 Models that Leave Behind a Surviving Companion for Normal Type Ia Supernovae</i> . RNAAS 10, 172.	[1]	2026
4. J. M. Pollin et al. <i>Multidimensional Nebular-Phase Calculations of Dynamically-Driven Double-Degenerate Double-Detonation Models for Type Ia Supernovae</i> . MNRAS 548, stag735.	[12]	2026
5. J. M. Pollin et al. <i>On the Fate of the Secondary White Dwarf in Double-Degenerate Double-Detonation Type Ia Supernovae – II. 3D Synthetic Observables</i> . MNRAS 533, 3036–3052.	[30]	2024
Co-authored (all include J. M. Pollin)	Citations	Year
6. F. P. Callan et al. <i>Multi-dimensional NLTE Radiative Transfer for a Double Detonation Type Ia Explosion Model in the Photospheric Phase</i> . MNRAS, in review.	[-]	2026
7. A. Holas et al. <i>Observational Signatures of Thermonuclear Electron-Capture Supernovae</i> . A&A 711, A172.	[-]	2026
8. R. Pakmor et al. <i>Violent Mergers Revisited: The Origin of the Fastest Stars in the Galaxy</i> . A&A 706, A239.	[11]	2026
9. A. Holas et al. <i>The Asymmetry of White Dwarf Double Detonations and the Observed Scatter around the Phillips Relation</i> . A&A 698, A269.	[7]	2025
10. F. P. Callan et al. <i>NLTE Spectral Modelling for a Carbon-Oxygen and Helium White Dwarf Merger as a Ca-Rich Transient Candidate</i> . A&A 702, A29.	[6]	2025
11. F. P. Callan et al. <i>Exploring the Range of Impacts of Helium in the Spectra of Double Detonation Models for Type Ia Supernovae</i> . MNRAS 539, 1404–1413.	[17]	2025
12. S. Srivastav et al. <i>Unprecedented Early Flux Excess in the Hybrid 02es-like Type Ia Supernova 2022ywc</i> . ApJ Lett. 956, L34. <i>Featured in Astrobites</i> .	[26]	2023
Software releases	Citations	Year
1. ARTIS Collaboration (incl. J. M. Pollin), <i>ARTISTOOLS</i> . Zenodo.	[3]	2023
2. ARTIS Collaboration (incl. J. M. Pollin), <i>ARTIS</i> . Zenodo.	[-]	2025